IMPERIAL COLLEGE LONDON

EE4-07
EE9-CS7-23
EE9-SO11

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected Copy

CODING THEORY

Wednesday, 1 May 10:00 am

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

All the questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : W. Dai
Second Marker(s) : C. Ling

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EE4-07 Coding Theory

Instructions for Candidates

Answer all three questions. Each question carries 25 marks.

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1. (Finite Fields)

(a) Let $a(x) = (x^2+1)\,b(x) \mod x^4+x+1$, where all the involved polynomials are in $\mathbb{F}_2[x]$. Suppose that $\deg(a(x)) = 0$. Find $a(x)$ and $b(x)$ by using Euclidean algorithm and Bézout's identity. [5]

(b) Let $\mathcal{C} \subset \mathbb{F}_2^7$ be a linear code. Its parity check matrix is given by

$$\boldsymbol{H}' = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

i Use Gaussian elimination to change the parity matrix into the form of $\boldsymbol{H} = [\boldsymbol{I}\ \boldsymbol{A}] \in \mathbb{F}_2^{3\times 7}$ where $\boldsymbol{I}$ is the identity matrix. [2]

ii Find the corresponding generator matrix $\boldsymbol{G} \in \mathbb{F}_2^{4\times 7}$ in the form $[\boldsymbol{B}\ \boldsymbol{I}]$. [2]

iii Assume that a message $\boldsymbol{m}_1$ is encoded into a codeword $\boldsymbol{c}_1$ using $\boldsymbol{G}$. The codeword $\boldsymbol{c}_1$ is transmitted over an binary symmetric channel. Let the received word be $\boldsymbol{y}_1 = [0\,0\,1\,1\,1\,1\,0]$. Compute the syndrome vector $\boldsymbol{s}_1$. Find the output of the minimum (Hamming) distance decoding, say $\hat{\boldsymbol{c}}_1$, and the corresponding transmitted message $\hat{\boldsymbol{m}}_1$. [3]

iv Assume that a message $\boldsymbol{m}_2$ is encoded into a codeword $\boldsymbol{c}_2$ using $\boldsymbol{G}$ The codeword $\boldsymbol{c}_2$ is transmitted over an binary erasure channel. Let the received word be $\boldsymbol{y}_2 = [?\,1\,1\,0\,?\,0\,1]$. Set the question marks in $\boldsymbol{y}_2$ to zero and compute the corresponding syndrome vector $\boldsymbol{s}_2$. Find the transmitted codeword $\boldsymbol{c}_2$ and the message $\boldsymbol{m}_2$. [3]

(c) Let $\mathcal{C} \subset \mathbb{F}_3^7$ be a linear code. (Note that $\mathbb{F}_3$ is used rather than $\mathbb{F}_2$.) Its parity check matrix is given by

$$\boldsymbol{H}' = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

i Use Gaussian elimination to change the parity matrix into the form of $\boldsymbol{H} = [\boldsymbol{I}\ \boldsymbol{A}] \in \mathbb{F}_3^{3\times 7}$ where $\boldsymbol{I}$ is the identity matrix. [2]

ii Find the corresponding generator matrix $\boldsymbol{G} \in \mathbb{F}_3^{4\times 7}$ in the form $[\boldsymbol{B}\ \boldsymbol{I}]$. [2]

(d)

i Find whether $x^2+1 \in \mathbb{F}_2[x]$ is irreducible or not. Prove your answer. [2]

ii Find whether $x^2 + 1 \in \mathbb{F}_3[x]$ is irreducible or not. Prove your answer. [2]

(e) Prove that in $\mathbb{F}_p[x]$, the equality that $(x+1)^p = x^p + 1$ holds.

Remark: You are allowed to use Euclid's Lemma: if integers a, b, c satisfy the properties that $a|(b \cdot c)$ and $\gcd(a, b) = 1$, then $a|c$. [2]

2. (Reed-Solomon codes)

Remark: You are allowed to use the facts that the distance of a linear code is the minimum number of the columns of its parity matrix $\boldsymbol{H}$ that are linearly dependent, and that the distance of a linear code is also the minimum number of nonzero elements in nonzero codewords.

(a) State and prove the Singleton bound for linear codes $\mathcal{C}[n,k,d]$. Codes that attain the singleton bound are maximum distance separable (MDS). [3]

(b) Define Reed-Solomon codes in the following way. Let $\mathbb{F}_q$ be a finite field and α be a primitive element of $\mathbb{F}_q$. Let $n = q-1$. For any polynomial $f(x) \in \mathbb{F}_q[x]$, define the evaluation map $\text{eval}(f(x))$ by mapping $f(x)$ to a vector $\boldsymbol{c} \in \mathbb{F}_q^{n-1}$:

$$f(x) \mapsto \boldsymbol{c} = [c_1, \cdots, c_n], \text{ where } c_i = f\left(\alpha^{i-1}\right).$$

An $[n,k]$ Reed-Solomon code is defined as

$$\mathcal{C}[n,k] = \{\text{eval}(f(x)),\ 0 \leq \deg f(x) \leq k-1\}.$$

i Prove that Reed-Solomon codes are linear codes. [3]

ii Prove that Reed-Solomon codes attain the Singleton bound. That is, Reed-Solomon codes are MDS. [3]

iii Based on the definition, find the corresponding generator matrix $\boldsymbol{G}$. [3]

iv The parity check matrix of a Reed-Solomon code can be written as

$$\boldsymbol{H} = \begin{bmatrix} 1 & \alpha & \cdots & \alpha^{n-1} \\ 1 & \alpha^2 & \cdots & \alpha^{2(n-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha^{n-k} & \cdots & \alpha^{(n-k)(n-1)} \end{bmatrix},$$

where the (j,i)-th entry is given by $h_{j,i} = \alpha^{(i-1)j}$, $1 \leq j \leq n-k$ and $1 \leq i \leq n$. Let $\boldsymbol{y} = \boldsymbol{c} + \boldsymbol{e}$ be the received word, where $\boldsymbol{c}$ is the transmitted codeword from a Reed-Solomon code and $\boldsymbol{e} = [e_1, \cdots, e_n]$ is the error vector introduced by the channel. Define the index set of errors by $\mathcal{I} = \{i : e_i \neq 0,\ 1 \leq i \leq n\}$. Define the syndrome vector $\boldsymbol{s} = \boldsymbol{y}\boldsymbol{H}^T = [s_1, \cdots, s_{n-k}]$. Give the general expression for s_j, $1 \leq j \leq n$, using the error symbols e_i, $i \in \mathcal{I}$. [2]

v Define the syndrome polynomial $S(z) = \sum_{j=0}^{n-k-1} s_{j+1} z^j$. Prove that

$$S(z) \equiv \sum_{i \in \mathcal{I}} \frac{e_i \alpha^{i-1}}{1 - \alpha^{i-1} z} \bmod z^{n-k}.$$

[5]

vi Define the error locator polynomial

$$L(z) = \prod_{i \in \mathcal{I}} \left(1 - \alpha^{i-1} z\right),$$

and the error evaluator polynomial

$$E(z) = \sum_{i \in \mathcal{I}} e_i \alpha^{i-1} \prod_{j \in \mathcal{I} \setminus \{i\}} \left(1 - \alpha^{j-1} z\right).$$

Specify the key equation used for Reed-Solomon decoding. [2]

vii Describe the process to use the syndrome polynomial $S(z)$ to find both the error locator polynomial $L(z)$ and the error evaluator polynomial $E(z)$. [4]

3. (Cryptography)

Remark: You are allowed to use the fact that for all nonzero elements $\alpha, \beta \in \mathbb{F}_q$, their product $\alpha \cdot \beta$ is also nonzero.

(a) Fermat's Little Theorem states that for every $\beta \in \mathbb{F}_q^*$, one has $\beta^{q-1} = 1$. It can be proved using the following arguments.

i For every $\beta \in \mathbb{F}_q^*$, define $\beta\mathbb{F}_q^* = \{\beta\beta_1, \cdots, \beta\beta_{q-1}\}$. Prove that $\beta\mathbb{F}_q^* = \mathbb{F}_q^*$. [2]

ii Prove that $\beta^{q-1} = 1$. [2]

(b) Prove that for every $\beta \in \mathbb{F}_q^*$, it holds that $\mathrm{ord}(\beta) \mid (q-1)$. [3]

(c) Consider the prime number $p = 311$.

i Find the missing values in the table below.

x	1	2	5	10	31	62	155
$2^x \bmod 311$	2	4	32	91	?	?	?
$17^x \bmod 311$	17	289	142	260	?	?	?

[6]

ii Between the numbers 2 and 17, which one is a primitive element in $\mathbb{F}_{311}$? Explain the reason. [4]

iii Find $\log_{17} 260 \bmod 311$ and $\log_{17} 36 \bmod 311$. [2]

(d) Consider RSA public key cryptography. Let the two involved prime numbers be $p = 311$ and $q = 313$.

i Let $(n, e) = (97343, 23)$ be the public key. Define $t = (p-1)(q-1)$. The private key d satisfies $e \cdot d = 1 \bmod t$. Find the corresponding private key d. [3]

ii Suppose that Alice would like to encrypt the message $m = 11$ using the public key (n, e). Find the encrypted word c. [3]